

Jeremy (Kai-Hsiang) Chang

📍 Montreal, CA

✉ Email

🐙 Github

🌐 LinkedIn

Education

McGill University

PhD in Computer Science

- Supervised by Jerome Waldispühl

Montreal, Canada

2024-Present

University of Guelph

M.Sc. in Microbiology

- Supervised by Julang Li

Ontario, Canada

2022-2024

University of Guelph - Ontario Veterinary College

B.Sc. in Biomedical Science

Montreal, Canada

2018-2022

Experience

McGill University

Graduate Research

Montreal, Canada

2024-Present

- Conducting research in **Antimicrobial Resistance Detection (AMR)** in **Metagenomics** and **Transcriptomics** with **Contrastive Learning** and **Large Language Models (LLMs)** using **Massive Multiplayer Online Games (MMOs)** as a source for high quality annotation.
- Work on newly developed **Adaptive Haar-like Distance (AHL D)** embedding pipeline for **Microbiome Biomarkers** for disease status.

Thyself

Backend Developer

Montreal, Canada (Remote)

2024-2025

- Worked on a startup project in a group of 15 people over 3 month period and resulted in **over 1,000 users**.
- Designed **Retrieval-Augmented Generation (RAG)** based **Article Recommendation System** with pre-trained LLMs. Developed backend service using **PostgreSQL** and a **FastAPI**.

McGill & Gearbox Softwares

Game Design/Research Intern

Montreal, Canada

2024-Present

- Design engaging minigame similar to **Borderland Science (BLS)** encoded with **biologically relevant data** for high quality annotation and **Contrastive Learning**.
- Direct collaboration with **Gearbox Softwares** for prototyping of the minigame using **Unity**, **Unreal Engine**, and **ReactJS (Three.js)** for UI and **AWS**, **FastAPI**, and **PostgreSQL** as a backend.

Animal Health Laboratory

Backend Developer

Guelph, Canada

2022-2023

- Participated in the design and development of an efficient internal inventory management system using **VBA**, **MySQL**, and internal API integrations. Successfully reduced workload of all technicians with 10 hours per week.

University of Guelph

Graduate Research Student

Guelph, Canada

2022-2023

- Apply and assess standard ML pipelines including **random forests**, **SVM**, **classic NN** on proteomics and metabolomics of urines to predict litter size and fertility status of pigs.
- Optimize feed fermentation process by analyzing high dimensional **time-series data** on temperature, moisture content, and pH level to reduce anti-nutritional factors.

Publications

Playing the Data: Video Games as a Tool to Annotate and Train Models on Large Datasets

Ghasemloo Gheidari, P., Chang, K.-H., Sarrazin-Gendron, R., Mutalova, R., Butyaev, A., Szantner, A., & Waldispühl, J.

ICML Workshop

2025

Influence of Bacillus licheniformis B4 fermentation on soybean meal nutritional value and early weaned pig growth performance.

Milmine, J., Fletcher, L., Chang, K. H., Kang, Y., Sun, M., Churcher, J., Jacobs, L., Song, Y., Patterson, R., Kiarie, E. G., & Li, J.

Journal of applied microbiology

2025

Skills

- ML Skills: Python, Pytorch, Sklearn, HuggingFace, Matplotlib, numpy, pandas
- Development & Tools: Linux, Git/Github, Docker, Django/Flask, FastAPI
- Frontend: Node.js, Three.js, React, Javascript/html/css, angular
- Programming Languages: Python, C, C++, C#, Java, R